

RIBBON-geographic-scaleup-roadmap

RIBBON — Manufacturing & Distribution Scale-Up Roadmap

⚠ **SUPERSEDED IN PART (Rev H.1).** This document predates Rev E (Level-as-Table / ribbon segments). Its freight assumptions (oversized monolith, ~class 400) and its distributed-microfactory recommendation no longer hold: stacked-segment freight (~class 85) strengthens the CENTRALIZED case — current guidance lives in PRD-12 and the crossover model, and the network decision waits on real quotes (Freight RFQ package). Phasing logic, DFMA principles, and make/buy/print reasoning remain useful history.

West → East → National → Global

Builds on the distributed-leaning hybrid from the productionization playbook. All figures are rough order-of-magnitude planning assumptions ($\pm 30\text{--}40\%$), not quotes.

Operating principle

Optimize **landed cost + lead time**, not ex-works fabrication cost. For an oversized, heavy, high-mix product, freight and lead time dominate — so:

- **Centralize** the flat, dense, standard, low-mix parts (base plates, spine, trough kits, hardware) at a few **parts hubs**; make them in volume and ship them compact.
- **Distribute** the bulky, custom work (top skin, core, insert kitting, final assembly, QA) to **regional microfactories** near demand, fed by the parametric file + a hub kit.
- **License** to add nodes fast once a region proves out. Never long-haul an assembled 3.9 m table — and never ship one across an ocean.
- The **digital thread** (one configurator → CAD/CAM → QA standard) is the connective tissue that keeps quality consistent across every node without central manufacturing.

Network roles

Role	Makes	Traits	Placement
Parts hub (centralized)	Base plates, steel spine, trough kits, hardware, power modules	Flat, dense, standard, low-mix — travels well	Few, central to the regions they feed
Regional microfactory (distributed)	Top skin (thermoform → LFAM), core, insert kitting, assembly, QA	Bulky, custom, final assembly — must be local	Many, near demand
Design/Ops HQ	Parametric engine, configurator, digital thread, QA + licensing	Asset-light IP core	One

Phased geographic roadmap

Phase	Timeframe	Nodes added	Made local vs. shipped in	Landed cost trend	Lead time	Primary risk
W — West	Yr 1-2	SoCal microfactory	Everything local at first;	Baseline	3-8 wks	Single point of

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anchor		(LA metro)	dense parts sourced regionally			failure
E — East node	Yr 2-3	Southeast microfactory + central parts hub (Midwest)	Coasts finish/assemble locally; hub ships compact dense parts to both	↓ (freight ~halved nationally)	3-6 wks	Two-node quality consistency
N — National mesh	Yr 3-4	+2-3 licensed microfactories (TX, Mid-Atlantic, +1)	Last-mile finishing local; LFAM tops begin at high-volume nodes	↓↓ (freight minimized)	1-3 wks	Partner onboarding & QA at scale
G — Global	Yr 4-6+	Continental nodes: Canada → Europe → APAC → MEA	Export the design , qualify regional fabricators; ship only specialized components	Region-local	In-region	Tariffs, standards, IP

Phase W — West anchor (Yr 1-2)

One SoCal finishing/assembly microfactory serving CA, the Pacific Northwest, Arizona, Nevada, and the mountain west. Dense parts are sourced regionally on an interim basis. Goal: prove the product, the configurator→CAM path, and the first references. Short-haul LTL keeps landed cost low across the West.

Phase E — East node → coast-to-coast (Yr 2-3)

Stand up a **Southeast microfactory** — recommended in the **Piedmont/Triad of North Carolina** (deep furniture-manufacturing cluster around High Point) or **Atlanta** (logistics depth + Savannah port) — serving the Eastern Seaboard, Southeast, and into the Mid-Atlantic/Northeast. Simultaneously open a **central parts hub in the Midwest metal-fab corridor** (Ohio/Indiana/Chicago) that makes base plates and spines in volume and ships them compact to *both* coasts, minimizing dense-part freight from a central point. Result: two finishing microfactories + one parts hub put ~90% of US institutional demand within medium-haul of a node, roughly halving national freight versus a single West node.

Phase N — National mesh (Yr 3-4)

Add 2-3 more regional microfactories to erase the last freight miles — Texas/South-Central (Dallas), Mid-Atlantic/Northeast (PA/NJ), and one more as volume warrants — onboarded via **licensing and on-demand networks** (regional solid-surface + metal shops, Xometry-type capacity) rather than owned plants. **Large-format additive (LFAM) tops** come online at the highest-volume nodes, eliminating mold cost and re-enabling true per-order geometry. Coverage becomes asset-light, fast (1-3 wk lead), and resilient.

Phase G — Global (Yr 4-6+)

Replicate make-near-demand internationally by **exporting the design, not the table**: qualify regional fabricators, stand up continental parts hubs, and ship only specialized components where needed. Sequence by attractiveness × ease of entry:

1. **Canada (Yr 4)** — adjacency and USMCA; serve from US nodes first, then a Toronto/Vancouver microfactory.
2. **Europe (Yr 4-5)** — strong education/institutional/design demand and sustainability alignment; metric-native. A Benelux/Germany microfactory + continental parts hub. **Our multi-height wave maps naturally onto EN 1729 educational-furniture size marks** — a credibility and spec advantage. Watch CE marking, accessibility norms, and ecodesign/EPR obligations; local content sidesteps EU furniture duties.
3. **APAC (Yr 5-6)** — fastest-growing contract-furniture region but the most competitive and manufacturing-dense. **License to strong regional manufacturers**, localize aggressively, and protect IP carefully; premium nodes (Singapore/ANZ) vs. cost partners (Vietnam/China).

4. **Middle East (Yr 5-6)** — hospitality and civic megaprojects; premium, partner-led.

Global considerations grid - Tariffs/trade — furniture carries meaningful import duties and trade actions; local make-near-demand avoids most of them, so distribution is a *tariff* strategy, not just a logistics one. - **Standards/certification** — US: BIFMA, Greenguard, ADA. EU: EN 1729 (education sizing), CE, accessibility, ecodesign/EPR. Each region needs local cert; the multi-height concept is a selling point against sizing standards. - **IP** — register design rights per jurisdiction; serialized files, controlled toolpaths, staged disclosure to licensees; strongest caution in low-enforcement markets. - **Supply resilience** — qualify multi-region suppliers for solid surface, steel, and stainless; prioritize recycled/low-carbon feedstock; nearshore where possible.

Make / buy / print — how it localizes

- **Top skin:** made-with-mold early in each new node → shifts to **printed (LFAM)** once that node's volume justifies it. Print capability rolls out node-by-node, west to east to global.
- **Dense standard parts** (base plates, spine, trough kits, hardware): consolidate to **continental parts hubs** — each region *buys from its hub*, not from overseas.
- **Inserts/lids:** printed at any new/low-volume node; tooled (bought) per continent once regional volume pays for molds.

When to open a node / centralize a part (rule of thumb)

- **Open a regional finishing node** when in-region annual volume × freight-saved-per-unit **exceeds** node standup + fixed cost — for an oversized product that tends to land around **~150-300 units/yr** in a region given the freight it saves.
- **Centralize a part** only when (per-unit scale saving × units) **exceeds** the added freight of shipping it in. Flat/dense parts clear this easily; the bulky top never does — which is the whole reason the top stays distributed.